

Myungje_Kim_Final_Project

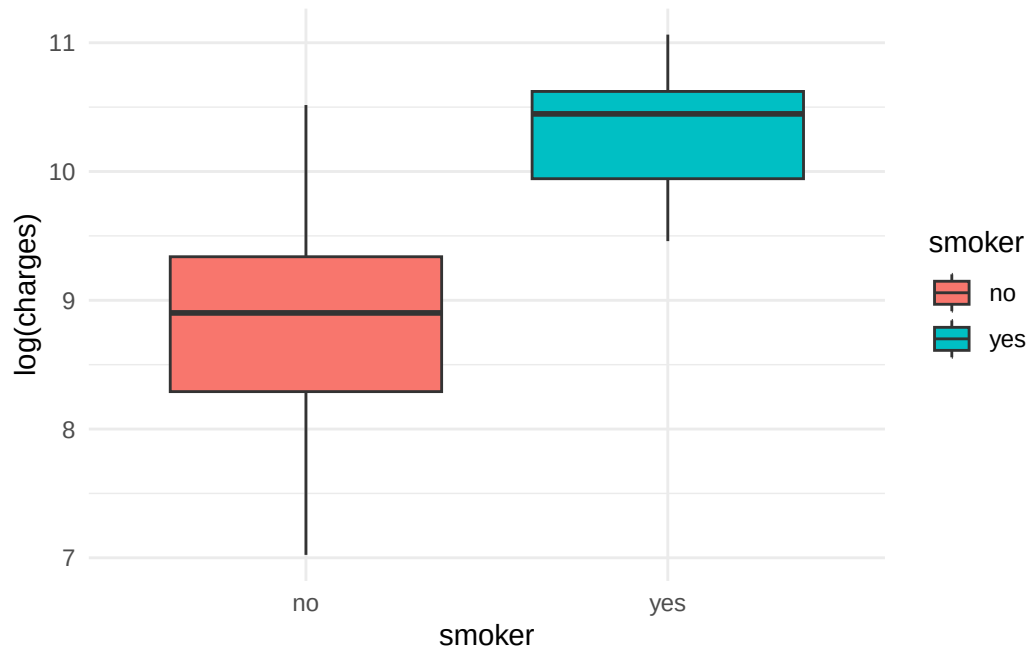
Myungje Kim

```
library(MASS)
library(tidyverse)
library(car)
library(Metrics)

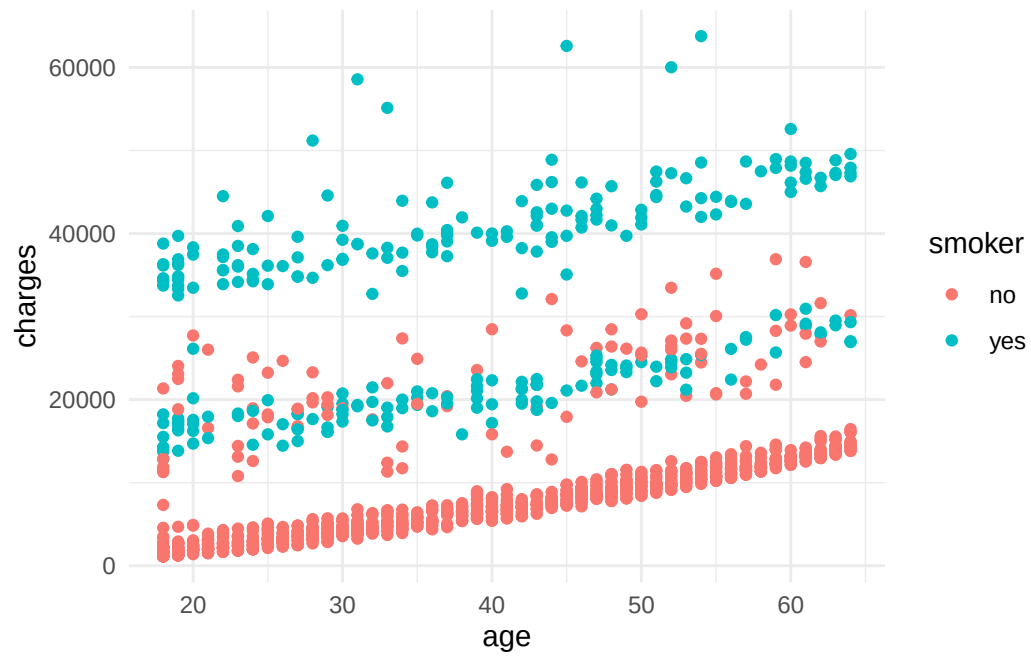
# Load data
ins <- read.csv("insurance.csv")
ins <- na.omit(ins)
ins <- ins |> mutate(across(c(sex, smoker, region), as.factor))
```

1. Exploratory Data Analysis

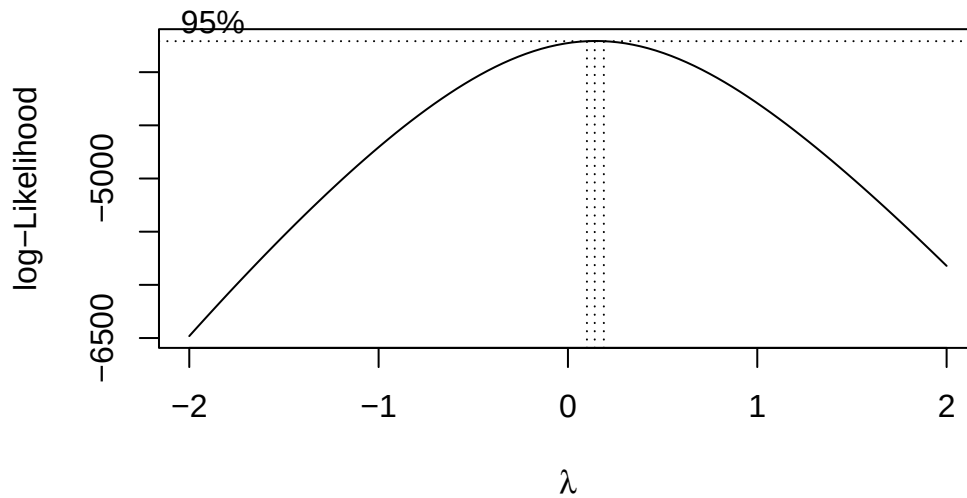
```
# Visualizing log-transformed relationships
ggplot(ins, aes(x = smoker, y = log(charges), fill = smoker)) + geom_boxplot() + theme_minimal()
```



```
ggplot(ins, aes(x = age, y = charges, color = smoker)) + geom_point() + theme_minimal()
```



```
# Box-Cox to justify log-transformation
boxcox(lm(charges ~ age + bmi + children + sex + smoker + region, data = ins))
```



2. Model Development

```
# Candidates
lm_full <- lm(log(charges) ~ age + sex + bmi + children + smoker + region, data = ins)

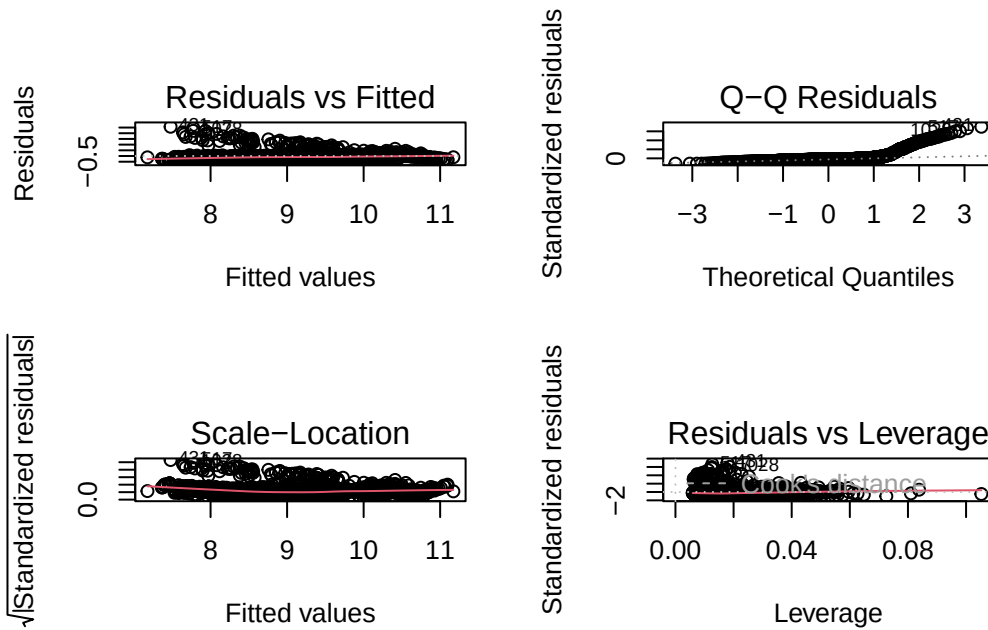
lm2 <- lm(log(charges) ~ age + I(age^2) + sex + bmi + I(bmi^2) + children +
          smoker + region + age:sex + age:children + age:smoker + age:region +
          sex:smoker + bmi:smoker + children:smoker + smoker:region, data = ins)

# Model selection using AIC
aic_results <- AIC(lm_full, lm2)
print(aic_results)
```

	df	AIC
lm_full	10	1637.033
lm2	24	1160.305

3. Model Diagnostics & Validation

```
# Diagnostics for the best model (lm2)
par(mfrow = c(2, 2))
plot(lm2)
```



```
par(mfrow = c(1, 1))

# Normality test
shapiro.test(residuals(lm2))
```

Shapiro-Wilk normality test

```
data: residuals(lm2)
W = 0.56769, p-value < 2.2e-16
```

```
# Cross-Validation
set.seed(999)
train_idx <- sample(1:nrow(ins), floor(0.7 * nrow(ins)))
train_data <- ins[train_idx, ]
test_data <- ins[-train_idx, ]
```

```
fit_train <- lm(formula(lm2), data = train_data)
preds <- exp(predict(fit_train, newdata = test_data))
rmse_val <- sqrt(mean((test_data$charges - preds)^2))
cat("Test RMSE:", rmse_val)
```

Test RMSE: 4905.941

4. Final Prediction Table

```
region_rep <- ins %>%
  group_by(region) %>%
  summarise(age = mean(age), bmi = mean(bmi), children = mean(children),
            sex = "male", smoker = "no")

preds_ci <- predict(lm2, newdata = region_rep, interval = "prediction", level = 0.95)

final_table <- ins %>%
  group_by(region) %>%
  summarise(Insurance_Charge = mean(charges), n = n()) %>%
  mutate(Lower_Limit = exp(preds_ci[, "lwr"]),
         Upper_Limit = exp(preds_ci[, "upr"]))

print(final_table)
```

```
# A tibble: 4 x 5
  region    Insurance_Charge      n Lower_Limit Upper_Limit
  <fct>          <dbl> <int>      <dbl>      <dbl>
1 northeast      13406.   324      3525.     15116.
2 northwest      12418.   325      3300.     14148.
3 southeast      14735.   364      3002.     12871.
4 southwest      12347.   325      2985.     12797.
```